

Finding the Fault in a Proof

Answer Key

High-school geometry

Quick Answers

- | | |
|-----------------------------------|------------------------------------|
| 1. — | 4. (b) the corrected value = 4, — |
| 2. (b) the corrected value = 8, — | 5. — |
| 3. — | 6. (b) the corrected value = 20, — |
-

What is verified

0 of 6 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 1, 2, 3, 4, 5, 6. The worked solution states what a correct response must contain.

Curriculum

Level: High-school geometry

HSA-REI.A.1 – *problem 2*

HSG-CO.C – *problems 1, 3, 4, 5, 6*

Difficulty 2–5 of 5 across 12 tagged checks.

Common wrong answers

2. *If they got 5: added 6 to the left side only, leaving the right side as $2x + 10$*
4. *If they got 2: subtracted 3 from both sides instead of adding it, so $3x = 6$*
6. *If they got 10: treated the linear pair as vertical angles and set the two measures equal*
-

1. Vertical angles proof at V : line 3’s reason is “because they are congruent”.

(a) **Line 3 is circular.** Its reason is the statement it is supposed to justify: the proof offers “ $\angle AVB \cong \angle CVD$ ” as the reason for “ $\angle AVB \cong \angle CVD$ ”. A reason has to come from outside the claim — from a given, a definition, a postulate, or an earlier line — so this step establishes nothing at all.

(b) Replace it with “**Vertical angles are congruent**” (the Vertical Angles Theorem). That reason traces back to line 1’s given and line 2’s definition, so the chain now runs forward from what was assumed to what is claimed.

Grading: (a) must identify line 3 *and* say the reason repeats the claim. (b) any reason that follows from the given, not from the conclusion.

2. Ari: $4x - 6 = 2x + 10$, concluding $x = 5$. (a) Find the error. (b) Correct value.

(a) **Line 2.** Going from $4x - 6 = 2x + 10$ to $4x = 2x + 10$ adds 6 to the left side only. The addition property of equality permits adding the same quantity to *both* sides, so the right side should have become $2x + 16$. The reason cited is the right property applied incorrectly, which is why the rest of the work looks tidy and is still wrong.

(b)

$$4x - 6 = 2x + 10$$

$$4x = 2x + 16$$

$$2x = 16$$

Check: $4(8) - 6 = 26$ and $2(8) + 10 = 26$.

$x = 8$

3. Interior ray $\overrightarrow{BD}$: line 2 claims $\angle ABD \cong \angle DBC$ “because the halves look equal”.

(a) **Line 2 is unsupported.** The only given is that $\overrightarrow{BD}$ lies in the interior of $\angle ABC$, and interior rays need not bisect. The reason cited is the appearance of the drawing, and a figure is not evidence: figures are not drawn to scale, and a proof must rest on givens, definitions, postulates, or lines already proved. Line 3 then inherits the fault, since it depends on line 2.

(b) Add to the given: $\overrightarrow{BD}$ bisects $\angle ABC$ — or mark the two smaller angles congruent in the figure. Line 2’s reason then becomes the **definition of an angle bisector**, and line 3’s “definition of congruent angles” is correct as it stands.

Grading: (a) must name line 2 and reject the figure as evidence. (b) must supply the bisector given (or its congruence marks) and name the definition.

4. Sam: M midpoint of $\overline{PQ}$, $PM = 5x - 3$, $MQ = 2x + 9$, concluding $x = 2$. (a) Error. (b) Correct value.

(a) **Line 3.** Lines 1 and 2 are fine: the definition of midpoint really does give $5x - 3 = 2x + 9$. But moving to $3x = 6$ subtracted 3 from both sides when the -3 had to be *added* to clear it, so the right-hand constant should be $9 + 3 = 12$, not $9 - 3 = 6$. The stated reason (subtraction property) is what was done; it is not what was needed.

(b)

$$5x - 3 = 2x + 9$$

$$3x - 3 = 9$$

$$3x = 12$$

Check: $PM = 5(4) - 3 = 17$ cm and $MQ = 2(4) + 9 = 17$ cm.

$x = 4$

5. “Corresponding angles congruent, so $j \parallel k$ — and they are congruent because $j \parallel k$.”

(a) The argument is **circular**: each of the two claims is offered as the reason for the other, and neither is ever tied to anything independent. Written as a chain it is “ P because Q , and Q because P ” — which is consistent with P and Q both being false, so it proves nothing.

(b) Two valid repairs, and you must choose one direction:

- **Given the angle measures** (measured or stated), conclude the corresponding angles are congruent, then conclude $j \parallel k$ by the *Converse* of the Corresponding Angles Postulate.
- **Given $j \parallel k$** , conclude the corresponding angles are congruent by the Corresponding Angles Postulate.

Either way the argument starts outside the pair of claims and runs one way only.

Grading: (a) must say each claim rests on the other. (b) must name a starting point that is not one of the two claims, and use the postulate or its converse in the matching direction.

6. Linear pair $\angle GKH$, $\angle HKJ$: line 2 claims the measures are equal “because vertical angles are congruent”. (a) Why not. (b) Correct x .

(a) Vertical angles are the *opposite* pair formed by two intersecting lines. $\angle GKH$ and $\angle HKJ$ are adjacent — they share the ray $\overrightarrow{KH}$ and their other sides, $\overrightarrow{KG}$ and $\overrightarrow{KJ}$, are opposite rays. That is the definition of a linear pair, not of a vertical pair, so the theorem cited does not apply and line 2 is unsupported. The correct relationship comes from the **Linear Pair Postulate**: the two angles are supplementary, so their measures add to 180.

(b)

$$(3x + 20) + 5x = 180$$

$$8x + 20 = 180$$

$$8x = 160$$

Check: $m\angle GKH = 3(20) + 20 = 80$ and $m\angle HKJ = 5(20) = 100$, and $80 + 100 = 180$.

$x = 20$

Grading: (a) must reject “vertical” on the adjacency/opposite-ray grounds *and* name the Linear Pair Postulate.